

Optimism of Training Error Rate

- Take a training set $\mathcal{T} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$, $\mathbf{x}_i \in \mathbb{R}^p \forall i$, with joint distribution F .
- Consider a new data point (x^0, y^0) .
- We work with the model

$$y = f(\mathbf{x}) + \varepsilon, \quad \mathbb{E}(\varepsilon) = 0, \quad \mathbb{E}(\varepsilon^2) = \sigma^2,$$

so that $\mathbb{E}y = f(\mathbf{x})$.

- We have $\hat{f}$ estimated. The generalization error is then:

$$\text{Err}_{\mathcal{T}} = \mathbb{E}_{x^0, y^0} [L(y^0, \hat{f}(x^0)) \mid \mathcal{T}]$$

- Averaging over training sets yields the *expected training error*:

$$\text{Err} = \mathbb{E}_{\mathcal{T}}(\text{Err}_{\mathcal{T}})$$

- Typically, the training error

$$\overline{\text{Err}} = \frac{1}{N} \sum_{i=1}^N L(y_i, \hat{f}(\mathbf{x}_i))$$

will be less than the true error $\text{Err}_{\mathcal{T}}$.

Ex: Linear regression model

- Consider $y_i = \mathbf{x}_i' \beta + \varepsilon_i$, $\forall i = 1, \dots, N$, with test data $\tilde{\mathcal{T}} = \{(\tilde{\mathbf{x}}_i, \tilde{y}_i)\}_{i=1}^N$.

$$R_{Tr}(\beta) = \sum_{i=1}^N (y_i - \mathbf{x}_i' \beta)^2 \qquad R_{Te}(\beta) = \sum_{i=1}^N (\tilde{y}_i - \tilde{\mathbf{x}}_i' \beta)^2$$

We know that

$$\hat{\beta} = (X'X)^{-1} X'Y = \arg \min_{\beta} R_{Tr}(\beta), \quad \text{where } X \in \mathbb{R}^{N \times p} \text{ is the design matrix.}$$

As a consequence:

$$R_{Tr}(\hat{\beta}) \leq R_{Tr}(\beta) = \frac{1}{N} \sum_{i=1}^N \varepsilon_i^2, \quad \text{where } \beta \text{ denotes the "true" value of the parameter.}$$

$$\Rightarrow \mathbb{E}[R_{Tr}(\hat{\beta})] \leq \frac{1}{N} \sum_{i=1}^N \mathbb{E}(\varepsilon_i^2) = \sigma^2. \quad (*)$$

For a given test observation, we have:

$$\begin{aligned} \tilde{y}_i - \tilde{\mathbf{x}}_i' \hat{\beta} &= \tilde{y}_i - \tilde{\mathbf{x}}_i'(\beta + \hat{\beta} - \beta) \\ &= \tilde{\varepsilon}_i + \tilde{\mathbf{x}}_i'(\beta - \hat{\beta}) \end{aligned}$$

$$\begin{aligned} \Rightarrow \mathbb{E}[(\tilde{y}_i - \tilde{\mathbf{x}}_i' \hat{\beta})^2 \mid \mathcal{T}] &= \mathbb{E}[(\tilde{\varepsilon}_i + \tilde{\mathbf{x}}_i'(\beta - \hat{\beta}))^2 \mid \mathcal{T}] \\ &= \sigma^2 + \mathbb{E}[(\tilde{\mathbf{x}}_i'(\beta - \hat{\beta}))^2 \mid \mathcal{T}] + 2 \mathbb{E}[\tilde{\varepsilon}_i \tilde{\mathbf{x}}_i'(\beta - \hat{\beta}) \mid \mathcal{T}] \\ &= \sigma^2 + \mathbb{E}[(\tilde{\mathbf{x}}_i'(\beta - \hat{\beta}))^2 \mid \mathcal{T}] \end{aligned}$$

(assuming ε_i independent of the training data)

$$\begin{aligned} \Rightarrow \mathbb{E}[(\tilde{y}_i - \tilde{\mathbf{x}}_i' \hat{\beta})^2 \mid \mathcal{T}] &\geq \sigma^2 \\ \Rightarrow \mathbb{E}(R_{Te}(\hat{\beta}) \mid \mathcal{T}) &\geq \sigma^2 \Rightarrow \mathbb{E}[\mathbb{E}(R_{Te}(\hat{\beta}) \mid \mathcal{T})] \geq \sigma^2 \\ \Rightarrow \mathbb{E}(R_{Te}(\hat{\beta})) &\geq \sigma^2 \end{aligned} \quad (**)$$

Combining (*) and (**), we get that

$$\mathbb{E}(R_{Te}(\hat{\beta})) \geq \mathbb{E}(R_{Tr}(\hat{\beta})).$$

Intuition: A fitting method typically adapts to the training data, and hence the training error will be an optimistic estimate of the generalization error.

In-sample error and optimism

Now, for each training point $\{\mathbf{x}_i\}_{i=1}^N$, consider observing N new response values $\{y_i^0\}_{i=1}^N$. The *in-sample error* is then:

$$\text{Err}_{in} = \frac{1}{N} \sum_{i=1}^N \mathbb{E}_{y^0} [L(y_i^0, \hat{f}(\mathbf{x}_i)) \mid \mathcal{T}]$$

The optimism is then:

$$op = \text{Err}_{in} - \overline{\text{Err}} \quad \left(\text{we expect } op > 0 \text{ as } \overline{\text{Err}} \text{ is biased downward} \right)$$

and the *average optimism*: $\omega = \mathbb{E}(op)$.

For instance, under a squared loss:

$$\text{Err}_{in} = \frac{1}{N} \sum_{i=1}^N \mathbb{E}_{y^0} [(y_i^0 - \hat{f}(\mathbf{x}_i))^2 \mid \mathcal{T}]$$

$$\begin{aligned}
\Rightarrow \mathbb{E}(\text{Err}_{in}) &= \frac{1}{N} \sum_{i=1}^N \mathbb{E}[(y_i^0 - \hat{f}(\mathbf{x}_i))^2] \\
&= \frac{1}{N} \sum_{i=1}^N \mathbb{E}[(\varepsilon_i + f(\mathbf{x}_i) - \hat{f}(\mathbf{x}_i))^2] \\
&= \sigma^2 + \frac{1}{N} \sum_{i=1}^N \mathbb{E}[(\hat{f}(\mathbf{x}_i) - f(\mathbf{x}_i))^2]
\end{aligned}$$

and:

$$\begin{aligned}
\overline{\text{Err}} &= \frac{1}{N} \sum_{i=1}^N (y_i - \hat{f}(\mathbf{x}_i))^2 \\
\Rightarrow \mathbb{E}(\overline{\text{Err}}) &= \frac{1}{N} \sum_{i=1}^N \mathbb{E}[(y_i - f(\mathbf{x}_i) + f(\mathbf{x}_i) - \hat{f}(\mathbf{x}_i))^2] \\
&= \sigma^2 + \frac{1}{N} \sum_{i=1}^N \mathbb{E}[(\hat{f}(\mathbf{x}_i) - f(\mathbf{x}_i))^2] - \frac{2}{N} \sum_{i=1}^N \mathbb{E}[(y_i - f(\mathbf{x}_i))(\hat{f}(\mathbf{x}_i) - f(\mathbf{x}_i))] \\
&= \sigma^2 + \frac{1}{N} \sum_{i=1}^N \mathbb{E}[(\hat{f}(\mathbf{x}_i) - f(\mathbf{x}_i))^2] - \frac{2}{N} \sum_{i=1}^N \text{Cov}(y_i, \hat{y}_i)
\end{aligned}$$

Meaning that:

$$\omega = \mathbb{E}(\text{Err}_{in}) - \mathbb{E}(\overline{\text{Err}}) = \frac{2}{N} \sum_{i=1}^N \text{Cov}(y_i, \hat{y}_i)$$

The amount by which $\overline{\text{Err}}$ underestimates the true error depends on how strongly y_i affects its own prediction, i.e. the harder we fit the data, the greater is $\text{Cov}(y_i, \hat{y}_i)$, thereby increasing optimism.

We then get that:

$$\mathbb{E}(\text{Err}_{in}) = \mathbb{E}(\overline{\text{Err}}) + \omega \quad \Leftrightarrow \quad \mathbb{E}(\text{Err}_{in}) = \mathbb{E}(\overline{\text{Err}}) + \frac{2}{N} \sum_{i=1}^N \text{Cov}(\hat{y}_i, y_i)$$

Linear fit

For a linear fit $Y = X\beta + \varepsilon$, we know that the fitted values are:

$$\hat{Y} = X\hat{\beta} = X(X'X)^{-1}X'Y = P_X Y, \quad \text{where } P_X \text{ is the hat matrix.}$$

Then $\hat{Y} = P_X X\beta + P_X \varepsilon = X\beta + P_X \varepsilon$, so that:

$$\text{Cov}(\hat{Y}, Y) = \text{Cov}(X\beta + P_X \varepsilon, X\beta + \varepsilon) = \text{Cov}(P_X \varepsilon, \varepsilon)$$

$$\text{Cov}(\hat{Y}, Y) = \sigma^2 P_X$$

Let h_{ii} denote the i -th diagonal element of P_X . Then $\text{Cov}(\hat{y}_i, y_i) = \sigma^2 h_{ii}$.
Hence:

$$\frac{2}{N} \sum_{i=1}^N \text{Cov}(\hat{y}_i, y_i) = \frac{2}{N} \sigma^2 \sum_{i=1}^N h_{ii} = \frac{2}{N} \sigma^2 \text{Tr}(P_X)$$

Assuming X is full rank, i.e. $\text{rank}(X) = p$, and because P_X is idempotent, it has p eigenvalues equal to 1, so that $\text{Tr}(P_X) = p$ and:

$$\frac{2}{N} \sum_{i=1}^N \text{Cov}(\hat{y}_i, y_i) = \frac{2p}{N} \sigma^2$$

Leading to:

$$\mathbb{E}(\text{Err}_{in}) = \mathbb{E}(\overline{\text{Err}}) + 2 \frac{p}{N} \sigma^2.$$

The optimism increases linearly with the number of inputs but decreases as the training sample size increases.

An obvious way to estimate the prediction error would be to estimate the optimism and then add it to the training error $\overline{\text{Err}}$.